Final Project Report

Title: Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

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Team Member:

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1. INTRODUCTION

1.1 Project Overview

This project utilizes Tableau to analyze and visualize housing market data. The focus is on identifying patterns in sale prices based on features such as renovation year, house condition, and basement area. Through interactive dashboards, the project aims to provide real estate stakeholders with tools for informed decision-making.

1.2 Purpose

The purpose of this project is to transform raw housing data into meaningful insights using Tableau's visual analytics. It aims to simplify complex datasets, support comparative analysis, and uncover hidden patterns that influence housing prices.

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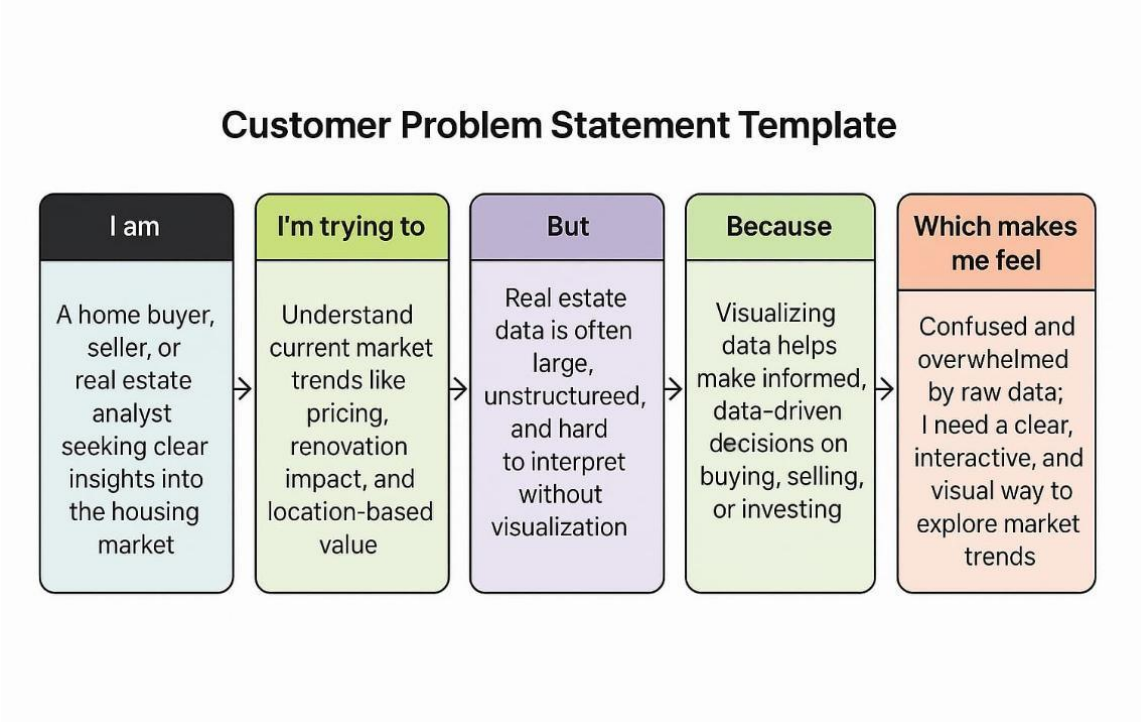
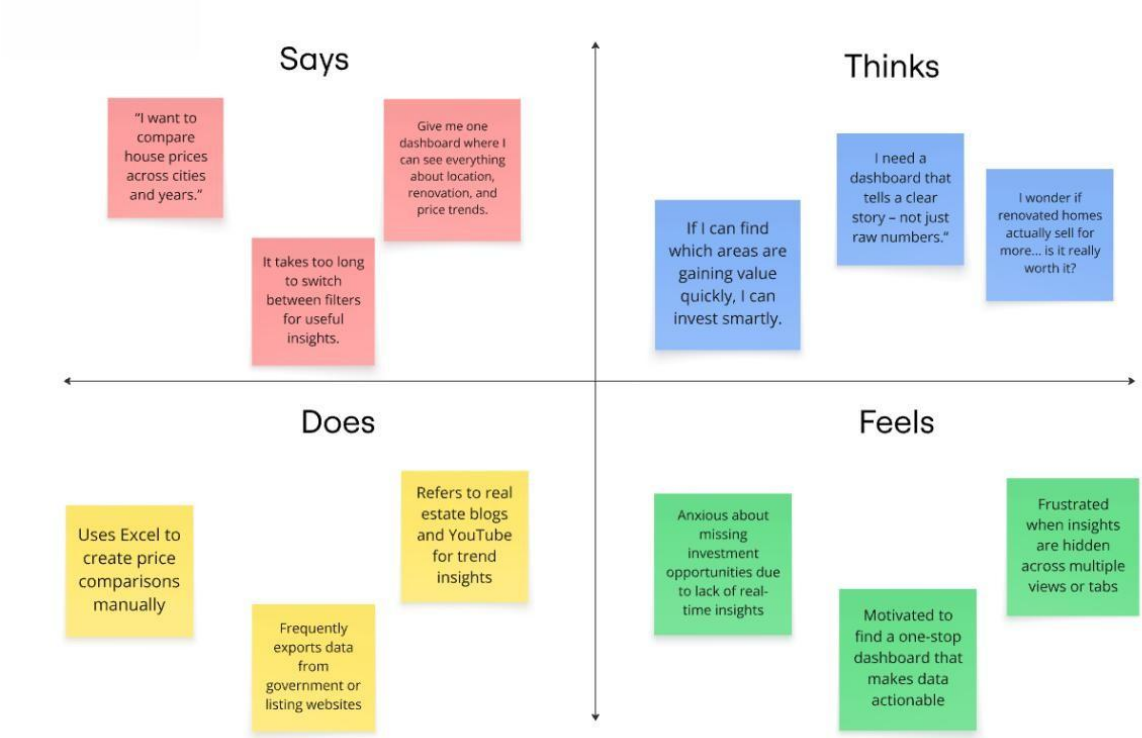
2. IDEATION PHASE

2.1 Problem Statement

The real estate market involves multiple factors influencing house prices, making it difficult for buyers, sellers, and investors to draw conclusions from raw data. This project addresses this challenge by offering visual insights into these influencing factors.



2.2 Empathy Map Canvas



2.3 Brainstorming

Key ideas and features discussed before implementation:

- Creating custom price bins (250K, 300K, …, 600K)
- Comparing price variation before and after renovations
- Evaluating the impact of house condition and basement area
- Calculating the age of houses based on the built year
- Creating a unified story by combining key dashboards in Tableau

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3. REQUIREMENT ANALYSIS

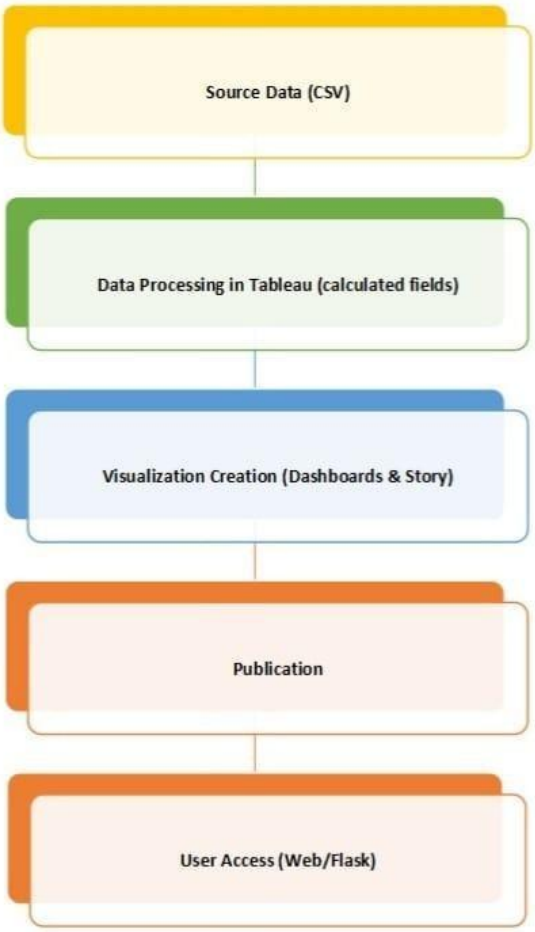
3.1 Customer Journey Map

Users begin by exploring price trends, use filters (like condition, area, renovation year), analyze charts for insights, and finally make informed decisions about property investments.

3.2 Solution Requirement

- Transformed dataset (CSV format from Kaggle)
- Tableau Desktop and Tableau Public for visualizations
- Optional: Python + Flask for embedding the dashboard into a web interface

3.3 Data Flow Diagram



3.4 Technology Stack

- **Visualization Tool:** Tableau
- **Dataset:** Kaggle - Transformed Housing Data
- **Backend :** Python, Flask

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4. PROJECT DESIGN

4.1 Problem-Solution Fit

The project simplifies raw data into visual formats, highlighting the relationship between property features and prices, thus offering an intuitive method to study the housing market.

4.2 Proposed Solution

- Create calculated fields (e.g., house age, sales price bins)
- Build multiple dashboards analyzing different variables
- Create a Tableau Story combining visual narratives
- Optionally integrate with a Flask-based user interface *4.3*

Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a user, I can collect housing data and load the data into Tableau.	1	High	ALL
Sprint-2	Data Preprocessing	USN-3	As a user, I can handle missing values in the dataset.	3	Medium	ALL

Sprint-1	Data Preprocessing	USN-4	As a user, I can encode and clean categorical data.	2	Medium	ALL
Sprint-3	Making Graphs/Visualizations	USN-5	As a user, I can build models based on processed data.	5	High	ALL
Sprint-4	Dashboard & Story	USN-6	As a user, I can design HTML interface for dashboard and story view.	6	High	ALL
Sprint-5	Report & Documentation	USN-7	Step by step guide documentation	7	Medium	ALL

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

S.No.	Parameter	Screenshot / Values
1	Data Rendered	Rendered from cleaned CSV containing ~21,000 housing records including Sale Price, Bedrooms, Bathrooms, Lot Area, Year Built, etc.
2	Data Preprocessing	Handled missing/null values, ensured correct data types (date, numeric), created calculated fields such as House Age.
3	Utilization of Filters	Applied filters in Tableau for Bedroom Count, House Condition, Renovation Year, House Age, and Price Range. Dashboard responsiveness under 2–3 seconds.
4	Calculation fields Used	- House Age - Total Sales by Renovation - Price per Square Foot - Sales by House Condition

5	Dashboard design	No of Visualizations/ Graphs - 1
6	Story Design	No of Visualizations/ Graphs – 1 story with 3 story points

- **Dashboard Load Time:** < 3 seconds (optimized by reducing unused fields)
- **Filter Responsiveness:** Filters like house condition, renovation year, and sale price bins responded without lag
- **Optimizations:** Used data extracts, avoided complex joins, applied filters only where necessary

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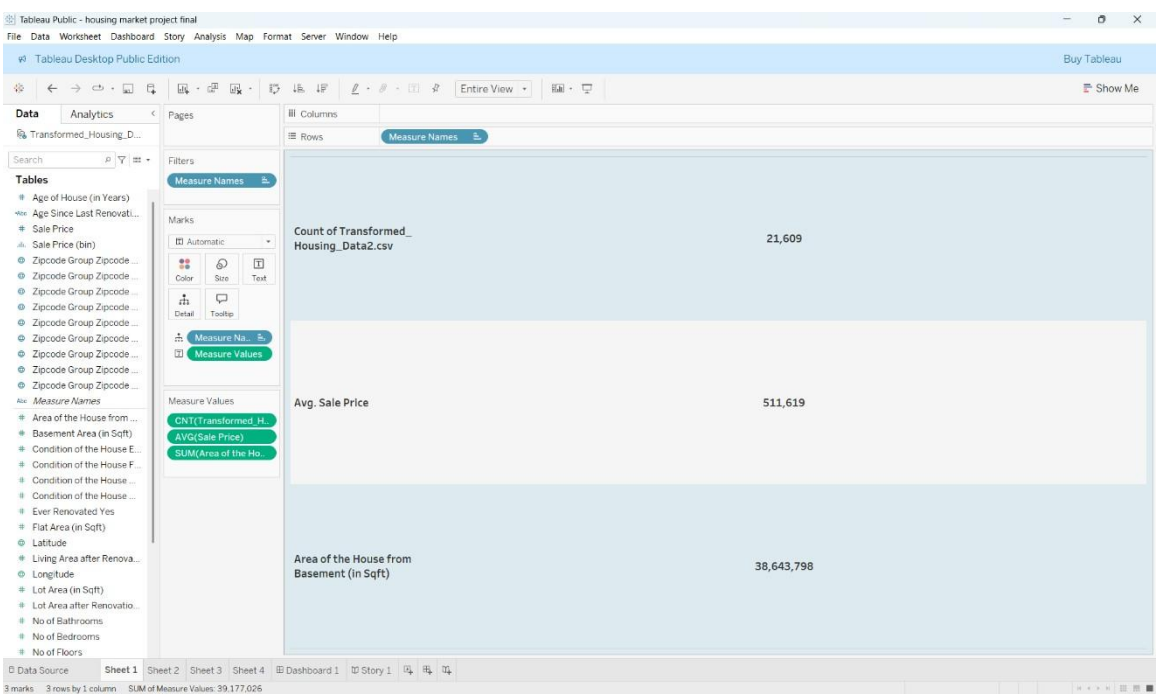
7. RESULTS

7.1 Output Screenshots

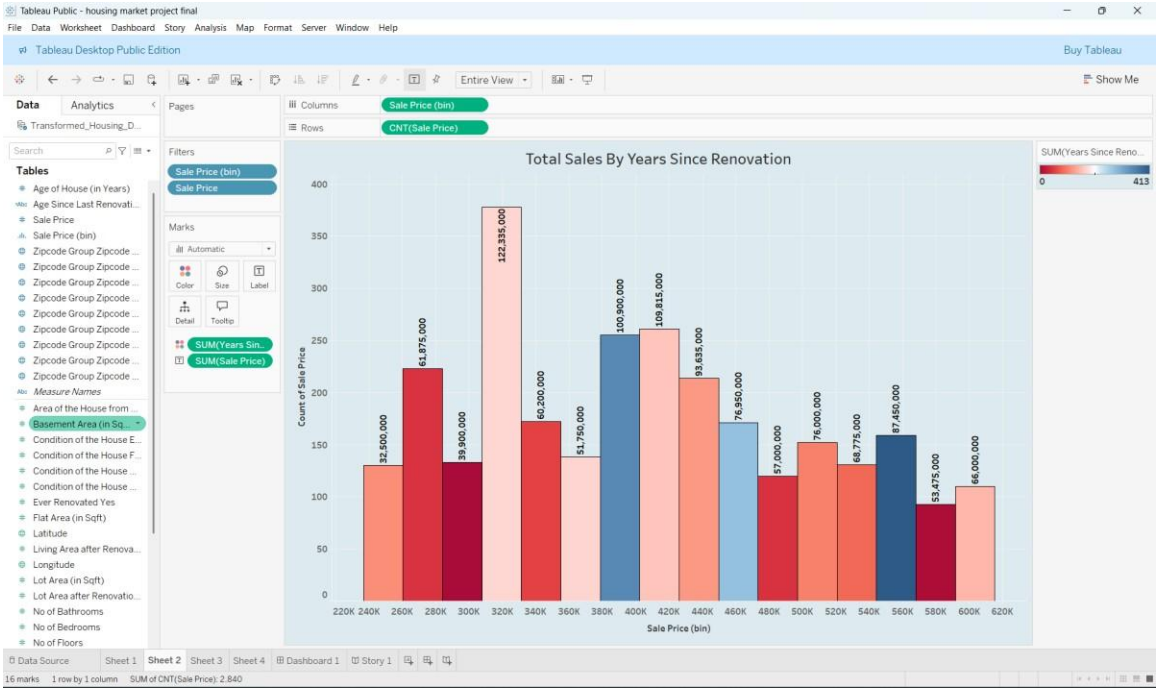
Count of Transformed_Housing_Data

Average Sale Prices

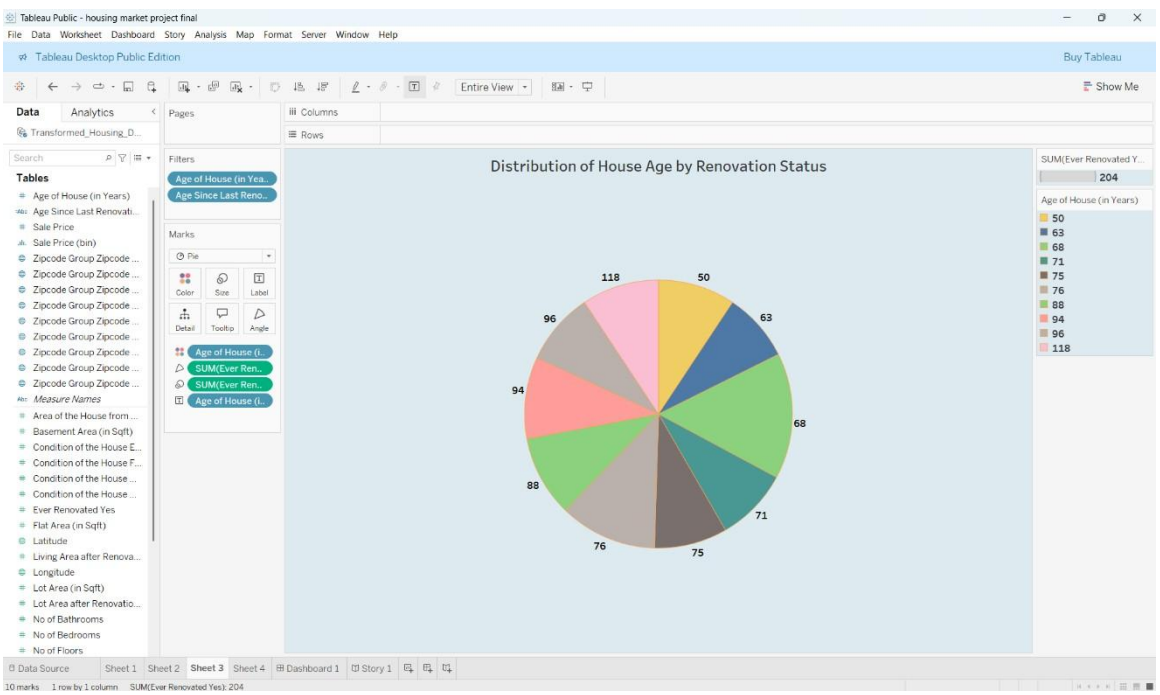
Area of House from Basement(in Sqft)



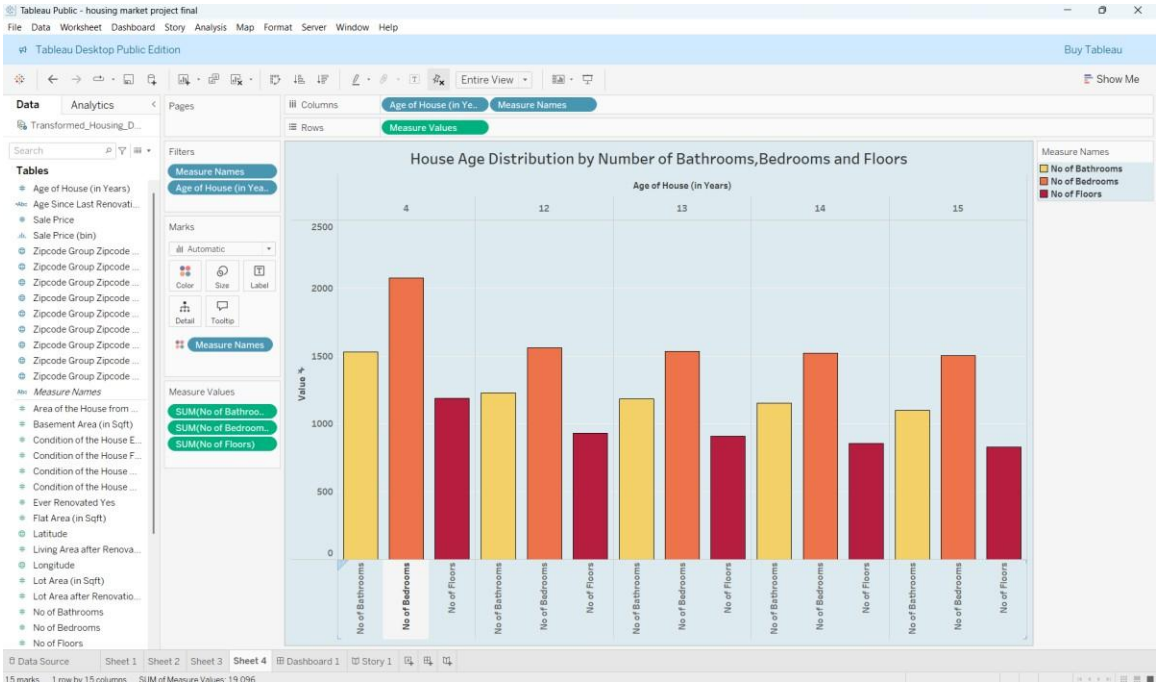
Total Sales by Years Since Renovation



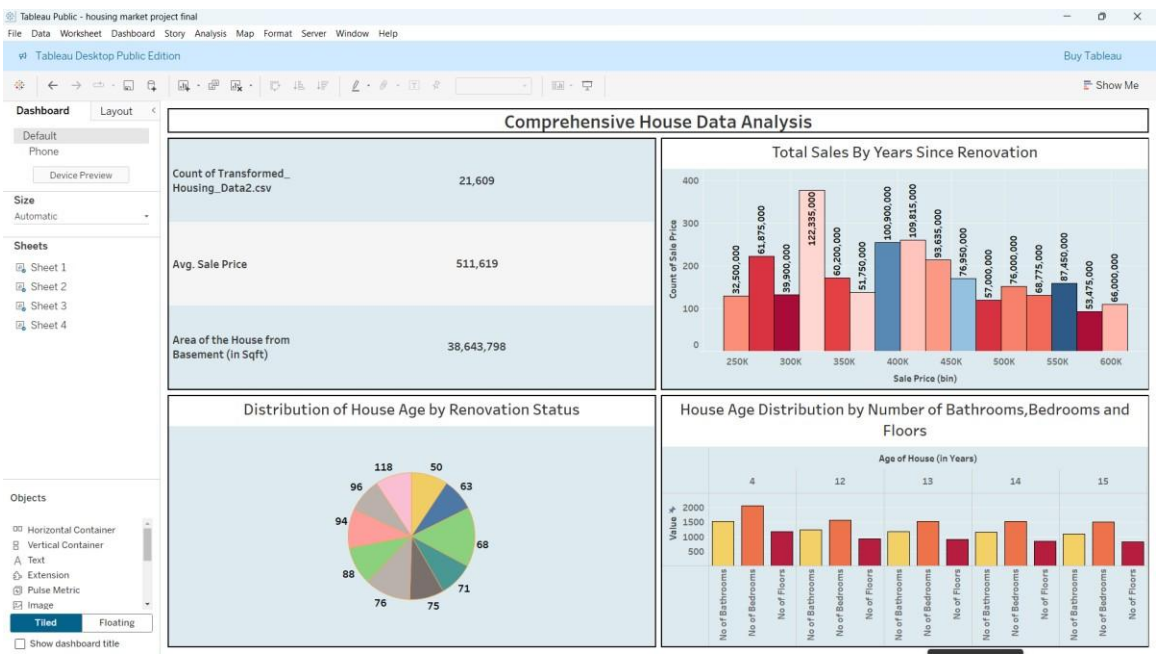
Distribution of House Age by Renovation Status



House Age Distribution by Number of Bathrooms, Bedrooms, and Floors



Dashboard



Story

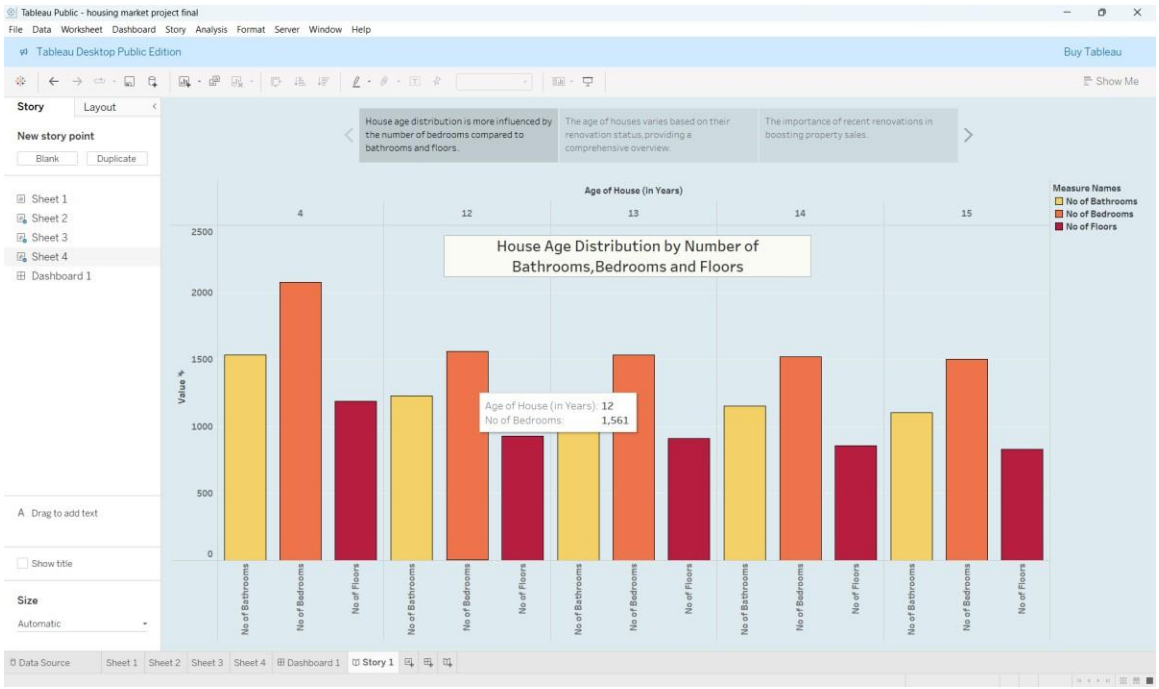
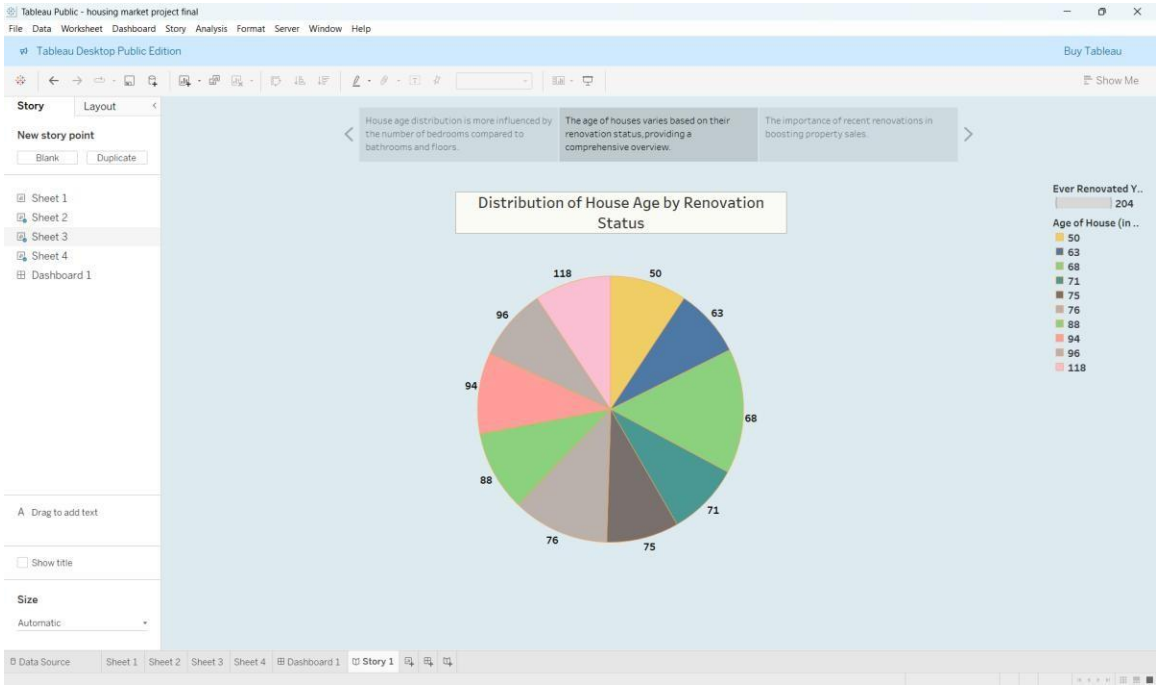


Tableau Public - housing market project final
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Dashboard 1
House age distribution is more influenced by the number of bedrooms compared to bathrooms and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovations in boosting property sales.
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
4
12
13
14
15
Value
2500
2000
1500
1000
500
0
Age of House (in Years): 12
No of Bedrooms: 1,561
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Drag to add text
Show title
Size
Automatic
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Dashboard 1
Story 1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Dashboard 1
House age distribution is more influenced by the number of bedrooms compared to bathrooms and floors.
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovations in boosting property sales.
Distribution of House Age by Renovation Status
50
63
68
71
75
76
88
94
96
118
Ever Renovated Y..
204
Age of House (in ..
50
63
68
71
75
76
88
94
96
118
Drag to add text
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Sheet 1
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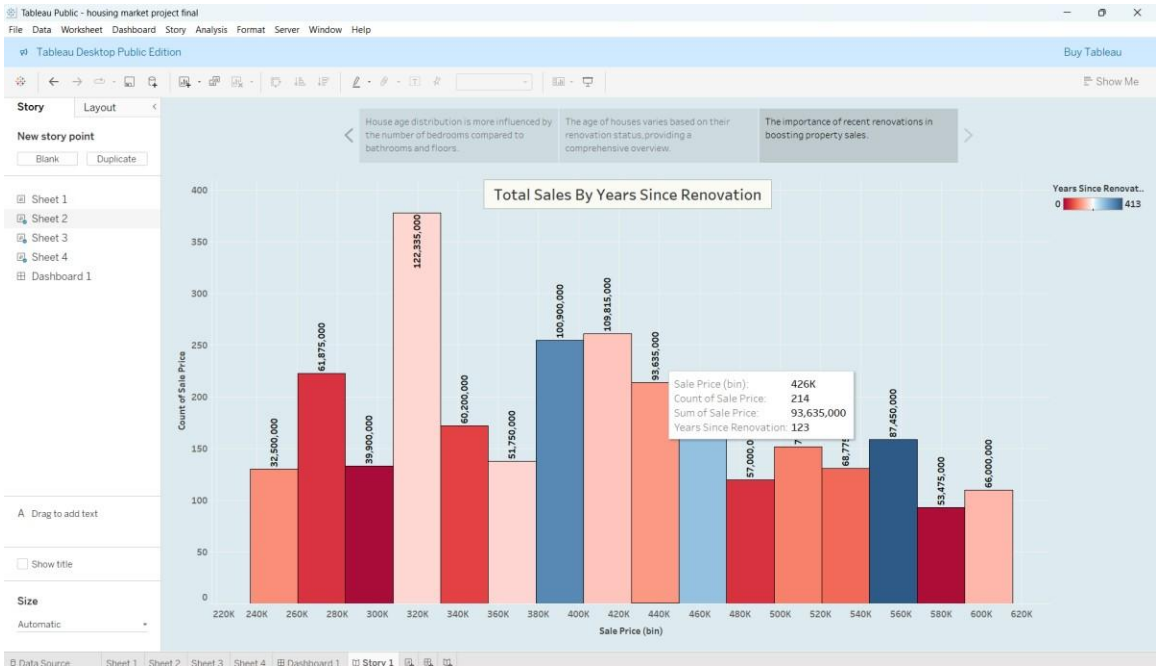

8. ADVANTAGES & DISADVANTAGES

Advantages

- Visual representation simplifies complex data analysis
- Enhances user interaction with filters and drill-downs
- Reusable dashboards for various stakeholders
- Web-embeddable for ease of access

Disadvantages

- Performance dependent on dataset size and design
- Public dashboards lack data confidentiality
- Limited machine learning integration without extra tools

9. CONCLUSION

The Tableau-based housing market analysis provides a clear and accessible way to interpret key trends and patterns in real estate data. It supports better decision-making

for buyers, sellers, and investors by highlighting critical sale price factors in a visually compelling format.

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10. FUTURE SCOPE

- Add predictive models using machine learning for price forecasting
- Integrate real-time APIs to fetch live market data
- Use geolocation mapping for regional trend analysis
- Provide user login functionality for personalized dashboards

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11. APPENDIX

- **Dataset:** https://www.kaggle.com/datasets/rituparnaghosh18/transformedhousing-data-2
- **Source Code**: https://github.com/hemanthkumar448/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau/tree/main/6.Project%20Execution%20Phase
- **GitHub Repo: https://github.com/hemanthkumar448/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau**